

Bachelor of Science in Business
Administration Undergraduate FALL Term
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**IT360: Information
Security**
IT 360 Project

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Sources : [Steganography Tutorial](#)
[springer.com](#)
[researchgate](#)
[scribd](#)
[dl.acm.org](#)

1. High-Level Design (HLD)

The proposed solution is a secure, end-to-end image steganography system designed to protect the existence of sensitive communication. It integrates **Encryption** with **Steganography** to provide two layers of protection.

Core Components

The system consists of four primary functional components:

- **Cover Medium:** The original digital image (JPEG or PNG) used to carry the hidden data.
- **Preprocessing Engine:** Converts the secret message into binary form and applies encryption.
- **Stego-Engine (Encoder/Decoder):** The central logic that embeds data into pixels or transform coefficients.
- **Security Controller:** Manages the **Stego Key**, which dictates how and where the data is hidden within the image.

Data Exchange Flow

Phase	Exchanged Data / Action	Description
Sender Input	Secret Message + Cover Image	The user provides the information to hide and the carrier.
Encoding	Message $\rightarrow$ Binary $\rightarrow$ Stego Image	The message is digitized and embedded via an algorithm (e.g., LSB or DCT).
Transmission	Stego Image	An image that looks visually unchanged is sent to the receiver.
Decoding	Stego Image + Key $\rightarrow$ Message	The receiver uses a specific key to extract and reconstruct the original data.

The interaction between the sender and receiver follows a structured functional process:

2. Proposed Tools & Technology Stack

The project identifies several tools and methods based on the domain of application:

- **Development Domain Tools:**
 - **Steghide:** For transform-domain embedding in compressed images (JPEG).
 - **StegoSuite / Xiao Steganography:** For spatial-domain embedding and user-friendly testing.
- **Evaluation Metrics:**
 - **PSNR (Peak Signal-to-Noise Ratio):** To measure technical noise levels.
 - **SSIM (Structural Similarity Index):** To ensure the stego image remains imperceptible to the human eye.
- **Programming Frameworks:** Implementation of adaptive methods (like **UNIWARD**) or AI-based detection requires machine learning models (CNNs or GANs).

3. Development Phases

The development of the IT 360 project is divided into four strategic phases:

Phase 1: Research & Requirements (Analysis)

- Define the balance between **Capacity**, **Robustness**, **Security**, and **Imperceptibility** based on the use case (e.g., humanitarian vs. general privacy).
- Select the target domain (Spatial vs. Transform).

Phase 2: Core Implementation (Development)

- **Encryption Layer:** Implement pre-embedding encryption to ensure that even if the data is detected, it remains unreadable.
- **Embedding Algorithm:** Develop the "Hiding Procedure," such as **LSB** for efficiency or **DCT** for resilience against image compression.

Phase 3: Steganalysis & Testing (Evaluation)

- Apply **Statistical Analysis** and **Visual Inspection** to the output to check for detectable anomalies.
- Test against "Cover Source Mismatch" to ensure the system works with images from various camera sensors.

Phase 4: Ethical Review & Finalization (Deployment)

- Assess the "Dual-Use" nature of the tool to ensure it aligns with positive applications, such as protecting activists or securing confidential communication.
- Finalize the technical report and user documentation.